

1 Acid–Base Imbalance & ABG Interpretation

Respiratory Acidosis – Causes

Definition: Happens due to ↓ ventilation → ↑ CO₂

بيحصل بسبب قلة التهوية → زيادة ثاني أكسيد الكربون: التعريف

Acute causes:

- Sedatives & anesthetics
→ المهدئات وأدوية التخدير
- Cardiac arrest
→ توقف القلب
- Airway obstruction (aspiration, laryngospasm)
→ انسداد مجرى الهواء
- Lung diseases (ARDS, severe asthma, pneumonia)
→ أمراض رئوية حادة

Chronic causes:

- COPD
→ الانسداد الرئوي المزمن
- Extreme obesity (Pickwickian syndrome)
→ السمنة المفرطة

Metabolic Acidosis – Causes

Definition: ↓ HCO₃⁻ or ↑ acids

نقص البيكربونات أو زيادة الأحماض: التعريف

- Diabetic ketoacidosis (DKA)
→ الحموضة الكيتونية

- Lactic acidosis
→ زيادة حمض اللاكتيك
 - Diarrhea (loss of HCO_3^-)
→ الإسهال
 - Kidney failure (AKI / CKD)
→ الفشل الكلوي
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Steps of ABG Interpretation

1. Check oxygenation (PaO_2 , SaO_2)
→ تقييم الأكسجة
 2. Check pH
 - < 7.35 = Acidosis
 - 7.45 = Alkalosis
→ تحديد الحموضة أو القلوية
 3. Find primary problem
 - PaCO_2 = respiratory
 - HCO_3^- = metabolic
→ نحدد السبب الأساسي
 4. Check compensation
 - Normal pH = full compensation
 - Abnormal pH = partial compensation
→ تعويض كامل أو جزئي
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2 Triage

Priority Levels

3 Levels:

- Emergent → Immediate
→ طوارئ فورية
- Urgent → Can wait ≤ 2 hours
→ عاجل
- Nonurgent → Stable
→ غير عاجل

4 Levels (Disaster):

- Red → Immediate
- Yellow → Urgent
- Green → Walking wounded
- Black → Dead / Expectant

5 Levels:

- Level 1: Resuscitation
- Level 2: Emergency
- Level 3: Urgent
- Level 4: Acute
- Level 5: Nonurgent

Types of Triage

- Simple triage → Mass casualties
→ فرز بسيط
 - Advanced triage → Limited resources
→ فرز متقدم
 - Reverse triage → Less injured first
→ العكس (حروب)
 - Telephone triage
→ فرز تليفوني
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Triage Systems

- Traffic director (Type I)
→ فحص سريع بواسطة غير متخصص
 - Spot checker (Type II)
→ أو طبيب RN
 - Comprehensive (Type III)
→ تقييم كامل مع إعادة تقييم
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3 ARDS

Definition

Severe lung injury with fluid in alveoli → ↓ oxygen exchange

متلازمة ضيق التنفس الحاد مع امتلاء الحويصلات بالسوائل →

Causes

Direct:

- Pneumonia, COVID-19

- Aspiration
- Chest trauma

Indirect:

- Sepsis
 - Shock
 - Pancreatitis
 - Drug overdose
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Nursing Management

- Monitor intake & output
→ متابعة السوائل
 - Daily weight
→ وزن يومي
 - Observe renal failure signs
→ متابعة الكلى
 - Turn patient every 2 hours
→ تغيير الوضعية
 - Provide rest
→ تقليل استهلاك الأكسجين
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4 Pulmonary Embolism (PE)

Venous Stasis Causes

- Immobility

- Atrial fibrillation
 - Vessel injury
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Causes of PE

- Thrombus from DVT
→ جلطة من الساق
 - Fat, air, tumor, amniotic fluid
→ مواد غير دموية
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Nursing Management

- Prevent DVT
→ تمارين + أجهزة ضغط
 - Monitor bleeding (anticoagulants)
→ متابعة النزيف
 - Improve oxygenation
→ وضعية وأكسجين
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5 Shock

Definition

Failure of tissue perfusion → cellular hypoxia

→ فشل وصول الدم للخلايا

Types

- Hypovolemic → ↓ volume
 - Cardiogenic → pump failure
 - Distributive → septic / anaphylactic / neurogenic
 - Obstructive → mechanical block
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Stages

1. Initial
 2. Compensatory
 3. Progressive
 4. Refractory (irreversible)
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Compensatory Stage

- Tachycardia
 - Vasoconstriction
 - ↑ contractility
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Progressive Stage

- Tissue hypoxia
 - Anaerobic metabolism
 - Metabolic acidosis
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Lab Findings

- ↑ Lactic acid
 - ↑ BUN & Creatinine
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Management Goals

- Maintain tissue perfusion
 - Secure airway
 - Restore cardiac output
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6 Chest Trauma

Rib Fracture Manifestations

- Severe pain
 - Bruising
 - Tenderness
 - Crepitus
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Flail Chest Complications

- Lung contusion
 - Atelectasis
 - Paradoxical breathing
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Tension Pneumothorax

- High-flow O₂

- Needle decompression (14G, 2nd ICS)
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Hemothorax Signs

- Hypotension
 - ↓ breath sounds
 - Dull percussion
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Cardiac Tamponade

- Pericardiocentesis
 - Thoracotomy
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Chest Tube Position

- Comfortable alignment
 - No tubing kinks
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ABG in Flail Chest

- ↓ PaO₂
 - ↑ PaCO₂
-

7 Respiratory Failure

Complications

- Brain hypoxia

- Dysrhythmias
 - DVT / PE
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Clinical Manifestations

- Restlessness
 - Confusion
 - Tachypnea
 - Cyanosis
 - Accessory muscles
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Causes

Extra-pulmonary:

- Brain injury, drugs, obesity

Intra-pulmonary:

- COPD, asthma, ARDS, PE
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8 Pain Management

Acute Pain

- Sudden
 - Short duration
 - Identifiable cause
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Types of Pain

- **Acute**
 - **Chronic**
 - **Nociceptive**
 - **Neuropathic**
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Non-Pharmacological

- **Massage**
 - **Heat / cold**
 - **Relaxation**
 - **Guided imagery**
 - **Music therapy**
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9 Cardiac Dysrhythmias

Ventricular Tachycardia

- **Rate > 100 bpm**
 - **Regular**
 - **No P waves**
 - **Wide QRS**
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Ventricular Fibrillation

- **CPR immediately**

- Defibrillation (200–360 J)
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Sinus Tachycardia Causes

- Exercise
 - Pain
 - Anxiety
 - HF
 - Dopamine / Atropine
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Atrial Flutter Causes

- CAD
 - Rheumatic disease
 - Cor-pulmonale
 - Alcoholism
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Atrial Fibrillation

Causes:

- HTN
- Valve disease
- CAD
- Pericarditis

Treatment:

- Rate control
 - Cardioversion
 - Anticoagulation
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10 Myocardial Infarction (MI)

Nursing Diagnoses

- Acute pain
 - ↓ Cardiac output
 - Activity intolerance
 - Anxiety
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Causes

- Atherosclerosis
 - Plaque rupture
 - Thrombosis
 - Coronary spasm
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Complications

- Dysrhythmias
- Heart failure
- Pulmonary edema
- Cardiogenic shock

Medical Management

- Morphine
- Nitroglycerin
- Streptokinase

1 1 Cardiac Surgery

Heart-Lung Machine

Takes over heart & lungs during surgery

→ يسمح بالجراحة والقلب ثابت

Pre-operative Care

- Stop anticoagulants
- NPO after midnight
- Chest shaving
- Antiseptic scrub

Post-operative Complications

- Bleeding
- Dysrhythmias
- Clots
- Stroke

- **Cardiac tamponade**

Nursing Diagnosis

- **Prevent complications**
- **Maximize comfort**